

A wholly nutritional 'multifocal angiostatic therapy' for control of disseminated cancer.



A great deal of effort is now being devoted to the development of new drugs that hopefully will control the spread of inoperable cancer by safely inhibiting tumor-evoked angiogenesis. However, there is growing evidence that certain practical nutritional measures have the potential to slow tumor angiogenesis, and it is reasonable to anticipate that, by combining several measures that work in distinct but complementary ways to impede the angiogenic process, a clinically useful 'multifocal angiostatic therapy' (MAT) might be devised. Several measures which might reasonably be included in such a protocol are discussed below, and include: a low-fat, low-glycemic index vegan diet, which may down-regulate the systemic IGF-I activity that supports angiogenesis; supplemental omega-3-rich fish oil, which has been shown to inhibit endothelial expression of Flk-1, a functionally crucial receptor for VEGF, and also can suppress tumor production of pro-angiogenic eicosanoids; high-dose selenium, which has recently been shown to inhibit tumor production of VEGF; green tea polyphenols, which can suppress endothelial responsiveness to both VEGF and fibroblast growth factor; and high-dose glycine, whose recently reported angiostatic activity may reflect inhibition of endothelial cell mitosis, possibly mediated by activation of glycine-gated chloride channels. In light of evidence that tumor-evoked angiogenesis has a high requirement for copper, copper depletion may have exceptional potential as an angiostatic measure, and is most efficiently achieved with the copper-chelating drug tetrathiomolybdate. If logistical difficulties make it difficult to acquire this experimental drug, high-dose zinc supplementation can achieve a slower depletion of the body's copper pool, and in any case can be used as maintenance therapy to maintain an adequate level of copper depletion. A provisional protocol is offered for a nutritionally based MAT entailing a vegan diet and supplemental intakes of fish oil, selenium, green tea polyphenols, glycine, and zinc. Inasmuch as cox-2 is overexpressed in many cancers, and cAMP can boost tumor production of various angiogenic factors as well as autogenous growth factors, adjunctive use of cox-2-specific NSAIDS may be warranted in some cases.